AI ASTROLOGER

Phase 8 — House Influence Propagation Engine

Complete Agent Roadmap · Weeks 29–30

Duration	Depends On	Provides To	Status
2 Weeks (29–30)	Phase 7 Complete	Phase 9 Temporal Engine	Ready to Build

1. Executive Summary

Phase 8 builds the House Influence Propagation Engine — the most mathematically advanced reasoning layer in the entire system. While Phases 5 through 7 analyse the chart as a collection of individual pieces (planet scores, yoga detections, aspect relationships), Phase 8 treats the chart as a connected network and models how influence flows across that network. The result is a ranked importance score for every house, revealing which life themes are most activated, most interconnected, and most prominent in a given chart.

At the end of Phase 7, the system knows which yogas are present and how strong each planet is. Phase 8 answers the next question: given all these planetary positions, aspects, and yoga activations, which houses emerge as the dominant themes of this person's life — and how do they influence each other? A chart where the 10th lord sits in the 7th, the 7th lord sits in the 10th, and a Raja Yoga connects both houses is not just a chart with a strong career and strong partnerships — it is a chart where career and partnership are deeply entangled, each feeding the other. The propagation engine surfaces this.

Why This Phase Is Critical

Without propagation, interpretation treats each house independently — the 10th house is analysed alone, the 7th house is analysed alone. With propagation, the engine models the network of connections and identifies emergent themes: 'Career and partnerships are the dominant life axes, with strong bidirectional influence between them.' This is what separates a mechanical interpretation from one that reflects the actual geometry of a chart.

2. What This Phase Builds

2.1 Primary Deliverable — propagation.py

A module that accepts a fully scored and yoga-detected chart (Phase 6 + Phase 7 output) and returns a ranked list of house importance scores, a set of dominant life theme identifications, and a map of inter-house influence connections with their strengths.

2.2 Secondary Deliverable — Updated Neo4j Schema

Each Chart node gains a HouseInfluence sub-graph: one HouseInfluence node per house storing its final importance score, its rank among the 12 houses, and the connections that contributed most to its score. These nodes are consumed by the Phase 9 Temporal Engine (which needs to know which houses matter most when evaluating dasha periods) and the Phase 11 NLG layer (which uses theme rankings to structure the interpretation narrative).

2.3 Tertiary Deliverable — Dominant Theme Identification

A post-propagation analysis step that maps the highest-scoring house clusters to named life themes (career, relationships, wealth, spirituality, health, etc.) and produces a ranked theme list for the chart. This becomes the skeleton of the Phase 11 interpretation output.

3. The Core Concept — Charts as Networks

The fundamental insight of Phase 8 is that a birth chart is not a static list of placements — it is a directed graph where influence flows along three types of edges. Understanding this network structure produces interpretive depth that no amount of isolated house or planet analysis can achieve.

3.1 The Three Connection Types That Create Edges

Connection Type	How It Creates an Edge	Edge Weight Basis
House Lordship	The lord of House A is placed in House B. This sends House A's themes into House B's territory.	Strength of the lord planet (Phase 6 score)
Planetary Aspect	A planet in House C aspects House D. This sends that planet's influence into House D.	Aspect strength (full 100% vs partial 75%) × planet score
Yoga Activation	A yoga connecting planets in Houses E and F creates a bidirectional reinforcement edge.	Yoga strength score (Phase 7 output)

A single chart typically generates 30–60 such edges across the 12 houses. The propagation engine traverses this edge network to compute how much cumulative influence is flowing into and through each house.

3.2 Why Isolated Analysis Fails

Consider a chart where: the 10th lord (Saturn) sits in the 7th house; the 7th lord (Venus) sits in the 10th house; Jupiter aspects both the 7th and 10th houses; and a Dhana Yoga connects the 2nd and 11th lords, both of whom aspect the 10th. An isolated analysis of the 10th house sees only Jupiter's aspect and perhaps Saturn's absence. The propagation model sees that the 10th house is receiving influence from the 7th (via Venus), from the 2nd (via yoga), from the 11th (via yoga), and has an outgoing edge to the 7th (via Saturn). Its importance score rises dramatically because it is both a major sender and receiver of chart influence.

4. The PageRank-Adapted Algorithm

The propagation algorithm is adapted from Google's PageRank model, which assigns importance to web pages based on how many other important pages link to them. In the chart context, a house is important if many other important houses direct influence into it. The adaptation requires three modifications from standard PageRank to fit the astrological domain.

4.1 Initialisation

Every house begins with an identical base importance score of 1.0. This neutral starting point means the algorithm does not pre-assume any house is inherently more important — importance emerges entirely from the chart's connection structure. The only exception is House 1 (the Lagna), which receives a fixed additional 0.3 seed value because classical astrology assigns the Lagna a foundational role in all chart matters.

4.2 The Iteration Step

On each iteration, every house's importance score is recalculated as a weighted sum of the scores of all houses currently sending influence to it, multiplied by the strength of each incoming edge. The formula per iteration is:

Variable	Definition
score(H) at iteration t+1	= damping_factor × Σ [score(sender) × edge_weight(sender→H)] + (1 − damping_factor) × base_score
damping_factor	Set to 0.85, consistent with standard PageRank (prevents score drainage to zero)
edge_weight(A→B)	Normalised strength of the connection from House A to House B, ranging 0.0 to 1.0
base_score	1.0 for all houses except House 1 (Lagna) which is 1.3

4.3 Convergence

The algorithm iterates until the maximum score change across all 12 houses between two consecutive iterations falls below a convergence threshold of 0.001. In practice this occurs within 15–25 iterations for typical charts. A hard cap of 100 iterations prevents infinite loops on pathological charts. The final scores after convergence are the raw importance values.

4.4 Normalisation

After convergence, scores are normalised so that the highest-scoring house receives exactly 10.0 and the lowest receives a proportionally scaled value. This normalised 0–10 scale is stored in Neo4j and used by all downstream phases. A parallel rank (1–12) is also stored.

Key Difference from Standard PageRank

Standard PageRank treats all outgoing links equally. The astrological adaptation uses weighted edges (strong planets create strong edges, weak planets create weak edges) and directional edges (house lordship is a one-way edge from the owned house to the placement house, not bidirectional). Yoga connections are the only edges that are inherently bidirectional because a yoga between two planets strengthens both houses simultaneously.

5. Edge Weight Computation — The Critical Step

The quality of the propagation output depends almost entirely on how accurately edge weights are computed. This is where the Phase 6 and Phase 7 outputs are consumed as inputs. Every edge in the house network has a weight between 0.0 and 1.0.

5.1 Lordship Edge Weights

When the lord of House A is placed in House B, the edge weight from A to B is determined by the lord planet's total strength score from Phase 6, normalised to the 0–1 scale (divide by 10). A lord with a strength score of 7.5 creates an edge weight of 0.75. A debilitated lord scoring 2.0 creates an edge weight of 0.20. This means that weak lords create weak lordship connections — the thematic link between the two houses exists, but it does not carry much influence.

5.2 Aspect Edge Weights

When a planet in House C aspects House D, the edge weight from C to D is the product of the planet's strength score (normalised) and the aspect's strength percentage from Phase 7. A Jupiter (strength 8.0) casting a full 100% 5th aspect creates an edge weight of 0.80. The same Jupiter casting a partial 75% 3rd aspect (if Saturn) would create an edge weight of 0.60. Sun and Moon do not create aspect edges on their own houses — only on the houses they aspect.

5.3 Yoga Edge Weights

Yoga edges are bidirectional. When a yoga connects planets in Houses E and F, an edge of equal weight is created from E to F and from F to E. The edge weight is the yoga's strength score from Phase 7 (already on a 0–1 scale). A strong Gaja Kesari Yoga at strength 0.82 creates edges of weight 0.82 in both directions between the houses occupied by Jupiter and Moon.

5.4 Edge Accumulation

Multiple edges can exist between the same pair of houses. For example, House 7 might send influence to House 10 via its lord's placement there, and additionally via a yoga connecting planets in both houses. These edges are summed and capped at 1.0 to prevent any single pair from dominating the network. The cap reflects the classical principle that there is a ceiling to how much two houses can reinforce each other.

6. Dominant Theme Identification

After propagation produces normalised house importance scores, a post-processing step identifies the dominant life themes and their relative prominence. This is the output that the Phase 11 NLG layer uses to structure the opening of any interpretation.

6.1 House-to-Theme Mapping

Each house maps to one or more life theme categories. The mapping comes from the ontology (houses.json) and is not recomputed here — Phase 8 only uses it to label the scores.

Houses	Primary Life Theme
1	Self, identity, constitution, personality
2	Accumulated wealth, family, speech, values
3	Effort, siblings, communication, courage, short journeys
4	Home, mother, property, emotional foundation, education
5	Creativity, children, intelligence, romance, past-life merit
6	Health, enemies, obstacles, service, daily work
7	Partnerships, marriage, public dealings, contracts
8	Transformation, hidden matters, inheritance, longevity
9	Fortune, dharma, father, higher learning, spirituality
10	Career, public status, authority, reputation
11	Gains, income, fulfilment of desires, social networks
12	Losses, liberation, foreign lands, spirituality, expenses

6.2 Theme Cluster Scoring

Some themes span multiple houses. Wealth, for example, is represented by Houses 2, 5, 9, and 11 in classical teaching. The theme cluster score is the average of the importance scores of all constituent houses. The top three theme clusters by score become the chart's dominant themes and are stored as ordered properties on the Chart node in Neo4j.

6.3 The Interpretation Skeleton

The dominant theme identification produces a ranked list in the following format, which is passed directly to Phase 11:

Field	Description
theme_rank	Integer 1–N, 1 = most dominant theme for this chart
theme_name	Canonical theme label (e.g. CAREER, PARTNERSHIPS, WEALTH)
theme_score	Float 0–10, averaged importance score of constituent houses
key_houses	List of house numbers most driving this theme's prominence
key_planets	List of planets most responsible for this theme's activation
yoga_contributors	List of yogas that boosted this theme's importance score

7. Neo4j Schema Updates

Phase 8 introduces two new node types to the knowledge graph and extends the Chart node with theme properties.

7.1 New Node Type — HouseInfluence

Property	Type & Description
house_number	Integer 1–12 — the house this node represents
raw_score	Float — score after propagation convergence, before normalisation
importance_score	Float 0–10 — normalised importance score
importance_rank	Integer 1–12 — rank among all 12 houses for this chart (1 = most important)
incoming_edges	JSON list — all edges sending influence to this house with weights
outgoing_edges	JSON list — all edges this house sends influence through
top_contributor	String — name of the single edge contributing most influence
theme_labels	String list — life themes this house is mapped to

7.2 New Node Type — DominantTheme

Property	Type & Description
theme_name	String — canonical theme name (CAREER, WEALTH, etc.)
theme_rank	Integer — rank among all themes for this chart
theme_score	Float 0–10 — cluster importance score
key_houses	Integer list — houses most contributing to this theme
key_planets	String list — planets most responsible for activation
yoga_contributors	String list — yoga names that boosted this theme

7.3 New Relationships

Relationship	Direction & Meaning
(Chart)-[:HAS_HOUSE_INFLUENCE]->(HouseInfluence)	Links chart to its 12 house importance nodes
(Chart)-[:HAS_DOMINANT_THEME]->(DominantTheme)	Links chart to its ranked theme nodes
(HouseInfluence)-[:INFLUENCED_BY]->(HouseInfluence)	Directed edge between two house nodes with weight property

8. Week-by-Week Task Plan

Week 29 — Edge Construction and Core Propagation

Day 1 — Interface Verification and Edge Builder Foundation

Before writing any propagation logic, verify that the Phase 7 outputs are available and correctly structured. Confirm that all DetectedYoga nodes are queryable by chart ID, all ChartPlanet nodes have strength_total populated, and all aspect relationships are stored as directed edges in Neo4j. Then build the edge builder module: a function that, given a chart ID, queries Neo4j and constructs the full weighted edge list for the house network. No propagation yet — only edge construction and validation.

Day 2 — Lordship and Aspect Edges

Implement lordship edge computation for all 12 houses. For each house, find its lord in the chart, retrieve the lord's strength score, and create a directed edge from the owned house to the placement house. Then implement aspect edges: for each planet, retrieve all its aspect

relationships from Phase 7, retrieve the planet's strength score, and create edges to all aspected houses weighted by aspect strength percentage. Test with 5 charts and manually verify the edge lists are correct.

Day 3 — Yoga Edges and Edge Accumulation

Implement yoga edge extraction: query all DetectedYoga nodes for the chart, identify the houses occupied by each yoga's participant planets, and create bidirectional edges weighted by yoga strength score. Then implement edge accumulation: where multiple edges exist between the same house pair, sum them and cap at 1.0. Validate the final edge list for 10 charts against manually computed expectations.

Day 4 — Propagation Algorithm Implementation

Implement the iterative propagation algorithm. Initialise all house scores to 1.0 with House 1 at 1.3. Run the iteration step with damping factor 0.85 until convergence (max change below 0.001) or 100 iterations. Apply normalisation to produce the 0–10 importance scores and 1–12 ranks. Store HouseInfluence nodes in Neo4j for 10 test charts. Verify that charts with known strong themes (e.g. a chart with clear career emphasis) produce high scores for the expected houses.

Day 5 — Dominant Theme Identification and Week 29 Integration

Implement the post-propagation theme clustering. Map house importance scores to theme clusters, compute cluster averages, rank themes, and create DominantTheme nodes in Neo4j. Run end-to-end on 20 test charts. Verify that the theme rankings make intuitive sense for charts with known configurations (e.g. a politician's chart should show career and authority as dominant themes).

Week 30 — Validation, Calibration, and Handoff

Day 1 — Expert Validation of Theme Rankings

Take 10 charts where the person's life themes are historically documented (public figures with known life narratives) and compare the propagation engine's dominant theme rankings against what an experienced astrologer would identify. Document discrepancies. Identify whether edge weights, damping factor, or theme cluster mappings need adjustment. This is a calibration day, not a code day — the goal is to understand whether the algorithm's output is directionally correct.

Day 2 — Algorithm Calibration

Based on Day 1 findings, adjust the parameters that need tuning. Likely adjustments include the damping factor (try 0.80 and 0.90 and compare), the Lagna seed value (try 1.5 instead of 1.3),

and the yoga edge bidirectionality (some traditions treat certain yogas as directionally asymmetric). Re-run the 10 expert charts with adjusted parameters and confirm improvement. Document the final parameter choices with rationale.

Day 3 — Full Suite Validation on 100 Test Charts

Run the propagation engine on all 100 test charts from Phase 4. Verify: every chart produces exactly 12 HouseInfluence nodes, importance scores are strictly between 0 and 10, ranks are strictly 1–12 with no ties (add tiebreaker logic if needed), the highest-scoring house is always a house that an astrologer would identify as prominent given the planet positions, and DominantTheme nodes are created and queryable for every chart. Log any charts that produce counter-intuitive results for investigation.

Day 4 — Phase 9 Interface Preparation

Define and document the query interface that Phase 9 will consume. Phase 9 needs to know: the importance score of any given house for the purpose of identifying which dasha periods will be most significant, the edge weight between any two houses for understanding how dasha activations propagate, and the dominant themes list for prioritising which predictions to generate first. Write the interface specification document and share with the Orchestrator for approval.

Day 5 — Handoff Preparation and Orchestrator Review

Finalise all Neo4j storage, run the complete success criteria checklist, and prepare the handoff package for Phase 9. The handoff package includes the propagation module's public query functions, the HouseInfluence and DominantTheme schema documentation, the calibration decisions log, and performance benchmarks. Notify the Orchestrator that Phase 8 is ready for review.

9. Success Criteria

Phase 8 is complete when all of the following criteria are met:

Criterion	Measurement	Pass Threshold
Edge construction completeness	All three edge types computed for every test chart	100% — no chart missing any edge type
Convergence reliability	Algorithm converges within iteration cap	100% of test charts converge
Score range validity	All importance scores within 0–10 range	Zero out-of-range values across 100 charts
Rank uniqueness	All 12 ranks distinct per chart	No ties — tiebreaker logic resolves all cases

Criterion	Measurement	Pass Threshold
Dominant theme validity	Top theme matches expert assessment	85%+ agreement on 10 expert-validated charts
Neo4j storage completeness	All HouseInfluence and DominantTheme nodes stored	Zero missing nodes across 100 charts
Phase 9 interface readiness	Query functions documented and tested	Orchestrator approves interface before Phase 9 begins
Performance	Propagation time per chart	Under 200ms per complete chart
Calibration documentation	Parameter choices recorded with rationale	All parameters documented before handoff

10. Integration Interfaces

10.1 Inputs Consumed by Phase 8

Input	Source & Description
ChartPlanet nodes with strength scores	Phase 6 — strength_total and component scores for all 9 planets
Aspect relationships in Neo4j	Phase 7 — directed ASPECTS relationships with strength percentages
DetectedYoga nodes	Phase 7 — yoga presence, strength scores, and planets involved
House lord assignments	Phase 4 — which planet rules which house for the chart's lagna
houses.json theme mappings	Phase 1 ontology — house-to-theme canonical mappings

10.2 Outputs Provided by Phase 8

Output	Consumed By
HouseInfluence nodes (12 per chart)	Phase 9 Temporal — identifies which houses matter most for dasha analysis; Phase 11 NLG — structures interpretation narrative
DominantTheme nodes per chart	Phase 9 Temporal — prioritises prediction generation; Phase 11 NLG — opening paragraph theme framing
Inter-house edge weights	Phase 9 Temporal — models how dasha activations propagate across the chart network

Output	Consumed By
Importance score ranking	Phase 11 NLG — determines which houses receive the most detailed interpretation

11. Risk Register

Risk	Likelihood	Impact	Mitigation
Weak charts produce flat score distributions (all houses similar)	Medium	Medium	Add a minimum spread threshold. If max score minus min score is below 2.0, flag the chart as 'low differentiation' rather than reporting misleading precision.
Yoga edges over-inflate certain houses	Medium	Medium	Cap bidirectional yoga edge weight at 0.8 maximum per pair. This prevents a single exceptional yoga from dominating the entire network score.
Convergence failures on highly interconnected charts	Low	High	Implement the 100-iteration hard cap and log any non-converging charts for manual inspection. Add a convergence diagnostic to the output object.
Phase 9 interface mismatch	Low	High	Define the interface contract at end of Week 29 Day 5 and get Orchestrator approval before Phase 9 begins.
Expert validation reveals systematic bias	Medium	High	Keep the calibration parameters as named constants from Day 1 so they can be adjusted without refactoring. Plan one full re-run of test charts after any parameter change.

12. Orchestrator Agent Notes

Architecture Compliance Requirements

The propagation engine must not recompute any value already produced by Phases 6 or 7 — it reads strength scores and yoga strengths through defined Neo4j queries only. Theme mappings must come from houses.json, not hard-coded in Python. All algorithm parameters (damping factor, convergence threshold, Lagna seed, edge cap) must be stored as named constants, not inline magic numbers. The HouseInfluence and DominantTheme node schemas must be approved by the Orchestrator before Neo4j writes begin.

Weekly reporting to Orchestrator should include: convergence statistics across test charts, dominant theme distributions (which themes appear most frequently as top-ranked), any charts that required manual inspection, and calibration parameter decisions with reasoning.

Handoff to Phase 9 is blocked until: Orchestrator approves the HouseInfluence and DominantTheme Neo4j schema, all success criteria are met, the Phase 9 interface contract is

documented and signed off, and expert validation results are reviewed. No Phase 9 development begins until this approval is received.